

SIMATS ENGINEERING

CO1

ASSESSMENT TOOL 1

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1.

Given

- File size = **150 MB = 150,000,000 bytes**
- Transport Layer segment size = **1500 bytes**
- Data Link Layer overhead = **40 bytes/frame**
- Network bandwidth = **50 Mbps = 50,000,000 bits/s**

1. Number of Transport Layer segments

[
 $\text{Number of segments} = \frac{150,000,000}{1500} = 100,000$
]

Total segments = 100,000

2. Number of Data Link Layer frames

Each segment is encapsulated into **one frame**.

Total frames = 100,000

3. Total Data Link Layer overhead

Overhead per frame = **40 bytes**

[
 $100,000 \times 40 = 4,000,000 \text{ bytes}$
]

Total overhead = 4,000,000 bytes (4 MB)

4. Total data transmitted

$$[150,000,000 + 4,000,000 = 154,000,000 \text{ bytes}]$$

Convert to bits:

$$[154,000,000 \times 8 = 1,232,000,000 \text{ bits}]$$

5. Transmission time

$$[\text{Transmission Time} = \frac{1,232,000,000}{50,000,000} = 24.64 \text{ seconds}]$$

Estimated transmission time = 24.64 seconds

(Ignoring propagation delay, processing delay, and retransmissions.)

Summary

Parameter	Value
File size	150 MB
Segment size	1500 bytes
Number of segments	100,000
Number of frames	100,000
Header/trailer per frame	40 bytes
Total Data Link overhead	4,000,000 bytes (4 MB)
Total transmitted data	154,000,000 bytes
Estimated transmission time	24.64 seconds

How the Transport and Data Link Layers ensure reliable communication

Transport Layer (Layer 4):

- Divides the file into segments.
- Assigns **sequence numbers** so data can be reassembled in the correct order.
- Uses **acknowledgments (ACKs)** to confirm successful delivery.
- Detects lost segments and performs **retransmissions**.
- Implements **flow control** to prevent the receiver from being overwhelmed.
- Uses **error detection** (checksums) to identify corrupted segments.

Data Link Layer (Layer 2):

- Encapsulates each segment into a frame by adding headers and trailers.
- Uses **Frame Check Sequence (FCS)/CRC** to detect transmission errors.
- Ensures correct delivery between **adjacent network devices**.
- May retransmit corrupted or lost frames on links that support **link-layer reliability**.
- Uses **MAC addressing** to deliver frames to the correct device on the local network.

Together, the **Transport Layer** provides **end-to-end reliability** between the client and server, while the **Data Link Layer** provides **node-to-node reliability** over each individual network link.

2.

Given

- Sliding window size = **6 frames**
- Data per frame = **1024 bytes**
- Total frames to transmit = **48 frames**

1. Number of transmission windows

$$\left[\begin{array}{l} \text{Number of windows} = \frac{48}{6} = 8 \end{array} \right]$$

Total transmission windows = 8

2. Total retransmissions

One frame is lost in **every window**.

- Number of windows = **8**
- Lost frames per window = **1**

$$\left[\begin{array}{l} 8 \times 1 = 8 \end{array} \right]$$

Total retransmissions = 8 frames

Final Answer

Parameter	Value
Total frames	48
Sliding window size	6 frames
Number of transmission windows	8
Lost frames	1 per window
Total retransmissions	8 frames

How flow control improves communication efficiency

Flow control is a mechanism used by the **Transport Layer** to ensure that the sender does not transmit data faster than the receiver can process it.

It improves communication efficiency by:

- Preventing the receiver's buffer from overflowing.
- Matching the sender's transmission rate to the receiver's processing speed.
- Reducing packet loss and unnecessary retransmissions.
- Improving network throughput by allowing continuous data transmission using the sliding window mechanism.
- Ensuring reliable and orderly delivery of data between sender and receiver.

Conclusion: Flow control increases the efficiency and reliability of data communication by balancing the data transmission rate between the sender and receiver.

3.

Given

- Original IP packet size = 12,000 bytes
- MTU = 1,500 bytes
- IP header per fragment = 20 bytes

1. Maximum data per fragment

Each fragment must fit within the MTU.

Maximum payload per fragment

$$1,500 - 20 = 1,480 \text{ bytes}$$

2. Number of fragments

The original packet contains 12,000 bytes of data.

Full-size fragments:

$$12,000 \div 1,480$$

8 remainder 160

So:

- 8 fragments carry 1,480 bytes each
- 1 final fragment carries 160 bytes

Total fragments

9

3. Total header overhead

Each fragment gets its own 20-byte IP header.

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Total IP header overhead

180 bytes

Fragmentation summary

Original packet size

12,000 bytes

MTU

1,500 bytes

Payload per full fragment

1,480 bytes

Number of fragments

9

Header per fragment

20 bytes

Total header overhead

180 bytes

How the Network Layer ensures successful delivery

The Network Layer (IP) is responsible for routing packets from the source to the destination and handling fragmentation when required.

1. Fragmentation

When a packet is larger than the MTU of a network link, the router:

- Splits the packet into smaller fragments.
- Adds an IP header to each fragment.
- Assigns the same Identification field to all fragments of the original packet.

2. Fragment Offset

Each fragment contains a Fragment Offset value indicating its position in the original packet. This allows the destination to place fragments in the correct order.

3. More Fragments (MF) Flag

- MF = 1 → more fragments follow.
- MF = 0 → this is the last fragment.

4. Reassembly at Destination

The destination host:

- Collects all fragments with the same Identification value.
- Uses the Fragment Offset to reorder them.
- Combines the payloads to reconstruct the original 12,000-byte packet.

5. Error Handling

If any fragment is lost:

- The complete packet cannot be reassembled.
- The Transport Layer (e.g., TCP) detects the missing data and requests retransmission.

Final Answer

Exam Ready

Item	Value
Number of fragments	9
Total IP header overhead	180 bytes

The Network Layer ensures successful delivery by fragmenting oversized packets, assigning identification numbers, using fragment offsets and MF flags, and enabling the destination host to reassemble all fragments into the original packet before passing it to the Transport Layer.

IP Fragmentation Breakdown

Data carried by each fragment when a 12,000-byte packet is fragmented with MTU 1,500 bytes.

Fragment	Data bytes
F1	1,480
F2	1,480
F3	1,480
F4	1,480
F5	1,480
F6	1,480
F7	1,480
F8	1,480
F9	160

4.

Given:

- Sliding window size = **6 frames**
- Data per frame = **1024 bytes**
- Total frames = **48**

Step 1: Calculate the number of transmission windows

[
 $\text{Number of windows} = \frac{48}{6} = 8$
]

Transmission windows required = 8

Step 2: Calculate the total retransmissions

One frame is lost in each window.

$$\begin{aligned} &[\\ &8 \times 1 = 8 \\ &] \end{aligned}$$

Total retransmissions = 8 frames

Final Answer

- **Sliding window size = 6 frames**
- **Total frames transmitted = 48**
- **Number of transmission windows = 8**
- **Total retransmissions = 8 frames**

How flow control improves communication efficiency

Flow control is a Transport Layer mechanism that regulates the rate of data transmission between the sender and receiver. It improves communication efficiency by:

- Preventing the receiver's buffer from overflowing.
- Matching the sender's transmission speed with the receiver's processing speed.
- Reducing packet loss and unnecessary retransmissions.
- Improving network throughput by allowing multiple frames to be sent before receiving acknowledgments.
- Ensuring reliable and efficient data transfer.

Thus, flow control increases the efficiency and reliability of communication by balancing the data flow between the sender and the receiver.

5.

Given:

- Video file size = **600 MB**
- Synchronization checkpoint = **every 60 MB**
- Failure occurs after **390 MB**

Step 1: Calculate the number of checkpoints created before failure

$$\begin{aligned} &[\\ &\frac{390}{60} = 6.5 \\ &] \end{aligned}$$

Only completed checkpoints are counted.

Checkpoints are created at:

- 60 MB
- 120 MB
- 180 MB
- 240 MB
- 300 MB
- 360 MB

Number of checkpoints before failure = 6

Step 2: Calculate the data to be retransmitted

The last checkpoint is at **360 MB**.

Failure occurs at **390 MB**.

[
 $390 - 360 = 30 \text{ MB}$
]

After recovery, transmission resumes from the last checkpoint.

Data to be retransmitted = 30 MB

Final Answer

- **Number of checkpoints created before failure = 6**
- **Data to be retransmitted after recovery = 30 MB**

Importance of synchronization in the Session Layer

The **Session Layer** manages communication sessions between devices and uses synchronization checkpoints to support recovery from failures. Its importance includes:

- Creating **checkpoints** during data transmission.
- Allowing communication to **resume from the last checkpoint** after a failure instead of restarting from the beginning.
- Reducing retransmission time and network overhead.
- Improving the reliability and efficiency of data transfer.
- Maintaining an organized and continuous communication session.

Thus, synchronization in the Session Layer ensures efficient recovery from failures by minimizing data retransmission and maintaining reliable communication.

6.

Given:

- Original file size = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%** of the compressed size

Step 1: Calculate the compressed file size

A 40% compression means the file size is reduced by 40%.

$$\begin{aligned} &[\\ \text{Compressed size} &= 900 - (40\% \times 900) \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ &= 900 - 360 = 540 \text{ MB} \\ &] \end{aligned}$$

Compressed file size = 540 MB

Step 2: Calculate the encrypted file size

Encryption adds **5% overhead** to the compressed size.

$$\begin{aligned} &[\\ 5\% \times 540 &= 27 \text{ MB} \\ &] \end{aligned}$$

$$\begin{aligned} &[\\ \text{Encrypted file size} &= 540 + 27 = 567 \text{ MB} \\ &] \end{aligned}$$

Encrypted file size = 567 MB

Step 3: Calculate the total percentage reduction

Reduction compared to the original file:

$$\begin{aligned} &[\\ 900 - 567 &= 333 \text{ MB} \\ &] \end{aligned}$$

Percentage reduction:

$$\left[\frac{333}{900} \times 100 = 37\% \right]$$

Total percentage reduction = 37%

Final Answer

- **Original file size = 900 MB**
- **Compressed file size = 540 MB**
- **Encrypted file size = 567 MB**
- **Total percentage reduction = 37%**

How the Presentation Layer improves transmission efficiency and security

The **Presentation Layer** is responsible for data formatting, compression, and encryption.

- **Compression** reduces the file size, decreasing transmission time and saving network bandwidth.
- **Encryption** protects the data from unauthorized access, ensuring confidentiality during transmission.
- It also performs **data translation**, allowing different systems to understand the transmitted data.

Thus, the Presentation Layer improves transmission efficiency by reducing data size and enhances security by encrypting the data before transmission.

7.

Given:

- Data size = **3 GB = 3000 MB**
- Throughput = **120 Mbps**
- Request size = **3 MB**

Step 1: Calculate the total number of requests

$$\left[\text{Number of requests} = \frac{3000 \text{ MB}}{3 \text{ MB}} = 1000 \right]$$

Total requests generated = 1000

Step 2: Calculate the total transmission time

Convert data size to megabits (Mb):

$$\begin{aligned} &[\\ &3000 \text{ MB} \times 8 = 24000 \text{ Mb} \\ &] \end{aligned}$$

Transmission time:

$$\begin{aligned} &[\\ &\text{Time} = \frac{24000 \text{ Mb}}{120 \text{ Mbps}} = 200 \text{ seconds} \\ &] \end{aligned}$$

Estimated transmission time = 200 seconds

Final Answer

- **Total data = 3 GB (3000 MB)**
- **Request size = 3 MB**
- **Total requests generated = 1000**
- **Effective throughput = 120 Mbps**
- **Estimated transmission time = 200 seconds**

How the Application Layer provides reliable services to users

The **Application Layer** is the topmost layer of the OSI model and provides network services directly to user applications.

- It enables services such as **FTP, HTTP, SMTP, and DNS**.
- It establishes communication between client and server applications.
- It divides data into requests for efficient transmission.
- It supports **authentication and user access control**.
- It detects and reports application-level errors.
- It ensures reliable file transfer using protocols like **FTP**, which confirm successful data transfer.

Thus, the Application Layer provides reliable services by enabling secure communication, managing user requests, and supporting dependable data transfer between applications.

8.

Given:

- Application Layer delay = **4 ms**
- Transport Layer delay = **3 ms**
- Network Layer delay = **5 ms**
- Data Link Layer delay = **2 ms**

- Physical Layer delay = **1 ms**
- Propagation delay = **18 ms**
- Transmission delay = **12 ms**

Step 1: Calculate the total processing delay

$$[4 + 3 + 5 + 2 + 1 = 15 \text{ ms}]$$

Total processing delay = 15 ms

Step 2: Calculate the total end-to-end delay

$$[\text{Total Delay} = \text{Processing Delay} + \text{Propagation Delay} + \text{Transmission Delay}]$$

$$[= 15 + 18 + 12 = 45 \text{ ms}]$$

Total end-to-end delay = 45 ms

Final Answer

- **Processing delay = 15 ms**
- **Propagation delay = 18 ms**
- **Transmission delay = 12 ms**
- **Total end-to-end delay = 45 ms**

Contribution of each OSI layer

- **Application Layer:** Provides network services to user applications (e.g., HTTP, FTP, Email).
- **Transport Layer:** Ensures reliable end-to-end communication through segmentation, flow control, and error recovery.
- **Network Layer:** Determines the best route and forwards packets to the destination using IP addressing.
- **Data Link Layer:** Encapsulates packets into frames, performs error detection, and controls access to the physical medium.
- **Physical Layer:** Transmits the actual bits over the communication medium using electrical, optical, or wireless signals.

Thus, all OSI layers work together to ensure efficient, reliable, and accurate communication, resulting in a total end-to-end delay of 45 ms.

9.

Given:

- Useful data = **80 MB = 80,000,000 bytes**
- Frame size = **1600 bytes**
- Protocol overhead per frame = **80 bytes**

Step 1: Calculate the useful data in each frame

$$\begin{aligned} & [\\ & \text{Useful data per frame} = 1600 - 80 = 1520 \text{ bytes} \\ &] \end{aligned}$$

Step 2: Calculate the total number of frames

$$\begin{aligned} & [\\ & \text{Number of frames} = \left\lceil \frac{80,000,000}{1520} \right\rceil \\ & = 52,632 \text{ frames} \\ &] \end{aligned}$$

Total frames required = 52,632

Step 3: Calculate the total overhead

$$\begin{aligned} & [\\ & 52,632 \times 80 = 4,210,560 \text{ bytes} \\ &] \end{aligned}$$

Total protocol overhead = 4,210,560 bytes (≈ 4.21 MB)

Step 4: Calculate the transmission efficiency

Total data transmitted:

$$\begin{aligned} & [\\ & 80,000,000 + 4,210,560 = 84,210,560 \text{ bytes} \\ &] \end{aligned}$$

Transmission efficiency:

$$\begin{aligned} & [\\ & \frac{80,000,000}{84,210,560} \times 100 \end{aligned}$$

= 94.999% \approx 95%
]

Transmission efficiency = 95%

Final Answer

- **Useful data per frame = 1520 bytes**
- **Total frames required = 52,632**
- **Total protocol overhead = 4,210,560 bytes (≈ 4.21 MB)**
- **Transmission efficiency = 95%**

How protocol overhead affects network performance

Protocol overhead consists of headers and trailers added by different OSI layers for addressing, routing, error detection, and control.

- It reduces the amount of useful data carried in each frame.
- It increases the total amount of data transmitted over the network.
- Higher overhead consumes more bandwidth and can increase transmission time.
- However, protocol overhead is essential because it provides reliable communication, error detection, flow control, and proper data delivery.

Thus, protocol overhead slightly reduces network efficiency but is necessary to ensure accurate, secure, and reliable data transmission.

10.

Given:

- **Original file size = 240 MB = 240,000,000 bytes**
- **Compression = 30%**
- **Segment size = 1400 bytes**
- **Data Link Layer overhead = 60 bytes per frame**

Step 1: Calculate the compressed file size

Compression reduces the file size by 30%.

[
\text{Compressed size} = 240 - (30\% \times 240)
]

[
= 240 - 72 = 168 \text{ MB}
]

Compressed file size = 168 MB

Convert to bytes:

$$[168 \times 1,000,000 = 168,000,000 \text{ bytes}]$$

Step 2: Calculate the total number of segments

$$[\text{Number of segments} = \frac{168,000,000}{1400} = 120,000]$$

Total segments transmitted = 120,000

Step 3: Calculate the total protocol overhead

$$[120,000 \times 60 = 7,200,000 \text{ bytes}]$$

Total protocol overhead = 7,200,000 bytes (7.2 MB)

Step 4: Calculate the final amount of data transmitted

$$[168,000,000 + 7,200,000 = 175,200,000 \text{ bytes}]$$

$$[= 175.2 \text{ MB}]$$

Final data transmitted across the Physical Layer = 175.2 MB

Final Answer

Parameter	Value
Compressed file size	168 MB
Number of segments	120,000
Total protocol overhead	7.2 MB (7,200,000 bytes)
Final data transmitted	175.2 MB

How multiple OSI layers ensure secure and reliable communication

- **Presentation Layer:** Compresses the medical record to reduce transmission time and may encrypt the data to protect patient privacy.
- **Transport Layer:** Divides the compressed data into segments, provides end-to-end reliability, flow control, and error recovery.
- **Network Layer:** Routes packets through the IoT healthcare network to the correct destination.
- **Data Link Layer:** Encapsulates segments into frames, adds headers/trailers, and performs error detection.
- **Physical Layer:** Transmits the final bits over the communication medium.

Thus, the OSI layers work together to provide efficient, secure, and reliable communication by compressing data, ensuring accurate delivery, detecting errors, and protecting sensitive medical information.